TF-IDF INFORMATION RETRIEVAL SYSTEM

GitHub link: Add your repository link after uploading.

Project: TF-IDF ranking using 10 different documents

Algorithm: Term Frequency-Inverse Document Frequency

Ranking method: Cosine Similarity

Includes Python code and sample output screenshots.

```
"""
TF-IDF Information Retrieval System — Different 10-Document Dataset
Uses only Python standard library.
Run: python ranking.py
"""
from pathlib import Path
from collections import Counter
import math
import re
import csv

BASE_DIR = Path(__file__).resolve().parent
DOCUMENTS_DIR = BASE_DIR / "documents"
OUTPUT_DIR = BASE_DIR / "output"
STOP_WORDS = {
    "a", "an", "and", "are", "as", "at", "be", "by", "can", "for",
    "from", "help", "in", "into", "is", "it", "of", "on", "or", "the",
    "their", "this", "to", "use", "uses", "using", "while", "with"
}

def tokenize(text):
    """Lowercase text and remove common stop words."""
    words = re.findall(r"[a-zA-Z]+", text.lower())
    return [word for word in words if word not in STOP_WORDS and len(word) > 1]

def load_documents():
    """Load all text files from documents/."""
    files = sorted(DOCUMENTS_DIR.glob("*.txt"))
    if not files:
        raise FileNotFoundError("No .txt files found in the documents folder.")
    return {f.name: f.read_text(encoding="utf-8") for f in files}

def build_tfidf(documents):
    """Calculate document term frequencies, IDF, and TF-IDF vectors."""
    tokens_by_doc = {name: tokenize(text) for name, text in documents.items()}
    df = Counter()
    for tokens in tokens_by_doc.values():
        df.update(set(tokens))
    n_docs = len(tokens_by_doc)
    idf = {term: math.log((1 + n_docs) / (1 + count)) + 1
           for term, count in df.items()}
    vectors = {}
```

```
    for name, tokens in tokens_by_doc.items():
        counts = Counter(tokens)
        total = len(tokens) or 1
        vectors[name] = {term: (count / total) * idf[term]
                         for term, count in counts.items()}
    return tokens_by_doc, vectors, idf

def cosine_similarity(a, b):
    """Return cosine similarity between sparse vectors."""
    dot = sum(a[t] * b[t] for t in (set(a) & set(b)))
    norm_a = math.sqrt(sum(v*v for v in a.values()))
    norm_b = math.sqrt(sum(v*v for v in b.values()))
    return dot / (norm_a * norm_b) if norm_a and norm_b else 0.0

def search(query, vectors, idf):
    """Weight query terms and rank documents by cosine similarity."""
    tokens = tokenize(query)
    counts = Counter(tokens)
    total = len(tokens) or 1
    query_vector = {term: (count / total) * idf[term]
                    for term, count in counts.items() if term in idf}
    ranked = [(name, cosine_similarity(query_vector, vector))
              for name, vector in vectors.items()]
    ranked.sort(key=lambda pair: (-pair[1], pair[0]))
    return tokens, ranked

def save_results(query, tokens, ranked):
    OUTPUT_DIR.mkdir(exist_ok=True)
    path = OUTPUT_DIR / "search_results.csv"
    with path.open("w", newline="", encoding="utf-8") as f:
        writer = csv.writer(f)
        writer.writerow(["query", "processed_query", "rank", "document", "cosine_similarity"])
        for rank, (name, score) in enumerate(ranked, 1):
            writer.writerow([query, " ".join(tokens), rank, name, f"{score:.6f}"])
    return path

def display_results(query, tokens, ranked):
    print("=" * 88)
    print("                TF-IDF INFORMATION RETRIEVAL SYSTEM")
    print("=" * 88)
    print(f"Query: {query}")
    print(f"Processed query: {' '.join(tokens) if tokens else '(no valid terms)'}")
    print("-" * 88)
```

```
    print(f"{'RANK':<6}{'DOCUMENT':<48}{'COSINE SIMILARITY':>20}")
    print("-" * 88)
    for rank, (name, score) in enumerate(ranked, 1):
        print(f"{rank:<6}{name:<48}{score:>20.6f}")
    if ranked and ranked[0][1] > 0:
        print("-" * 88)
        print(f"Top result: {ranked[0][0]} | Score: {ranked[0][1]:.6f}")
    else:
        print("No matching terms were found.")
    print("=" * 88)

def main():
    documents = load_documents()
    _, vectors, idf = build_tfidf(documents)
    print(f"Loaded {len(documents)} different documents.")
    while True:
        try:
            query = input("\\nEnter search query (or 'exit'): ").strip()
        except (EOFError, KeyboardInterrupt):
            print("\\nExiting.")
            break
        if query.lower() in {"exit", "quit"}:
            break
        if not query:
            print("Please enter a query.")
            continue
        tokens, ranked = search(query, vectors, idf)
        display_results(query, tokens, ranked)
        print(f"Results saved to: {save_results(query, tokens, ranked)}")

if __name__ == "__main__":
    main()
```

```
================================================================================
                TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: cybersecurity threat
Processed query: cybersecurity threat
--------------------------------------------------------------------------------
RANK  DOCUMENT                                        COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D01_Cybersecurity_Threat_Detection.txt                    0.242552
2     D02_Renewable_Wind_Energy.txt                             0.000000
3     D03_Water_Quality_Monitoring.txt                          0.000000
4     D04_Supply_Chain_Optimization.txt                         0.000000
5     D05_Speech_Emotion_Recognition.txt                        0.000000
6     D06_Plant_Disease_Classification.txt                      0.000000
7     D07_Electric_Vehicle_Battery_Health.txt                   0.000000
8     D08_Disaster_Response_Routing.txt                         0.000000
9     D09_Digital_Banking_Authentication.txt                    0.000000
10    D10_Air_Pollution_Forecasting.txt                         0.000000
--------------------------------------------------------------------------------
Top result: D01_Cybersecurity_Threat_Detection.txt | Score: 0.242552
================================================================================
```

```
================================================================================
                TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: renewable wind energy
Processed query: renewable wind energy
--------------------------------------------------------------------------------
RANK  DOCUMENT                                        COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D02_Renewable_Wind_Energy.txt                          0.503832
2     D01_Cybersecurity_Threat_Detection.txt                 0.000000
3     D03_Water_Quality_Monitoring.txt                       0.000000
4     D04_Supply_Chain_Optimization.txt                      0.000000
5     D05_Speech_Emotion_Recognition.txt                     0.000000
6     D06_Plant_Disease_Classification.txt                   0.000000
7     D07_Electric_Vehicle_Battery_Health.txt                0.000000
8     D08_Disaster_Response_Routing.txt                      0.000000
9     D09_Digital_Banking_Authentication.txt                 0.000000
10    D10_Air_Pollution_Forecasting.txt                      0.000000
--------------------------------------------------------------------------------
Top result: D02_Renewable_Wind_Energy.txt | Score: 0.503832
================================================================================
```

```
================================================================================
                TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: water quality
Processed query: water quality
--------------------------------------------------------------------------------
RANK  DOCUMENT                                        COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D03_Water_Quality_Monitoring.txt                          0.482761
2     D10_Air_Pollution_Forecasting.txt                         0.079036
3     D06_Plant_Disease_Classification.txt                      0.074440
4     D01_Cybersecurity_Threat_Detection.txt                    0.000000
5     D02_Renewable_Wind_Energy.txt                             0.000000
6     D04_Supply_Chain_Optimization.txt                         0.000000
7     D05_Speech_Emotion_Recognition.txt                        0.000000
8     D07_Electric_Vehicle_Battery_Health.txt                   0.000000
9     D08_Disaster_Response_Routing.txt                         0.000000
10    D09_Digital_Banking_Authentication.txt                    0.000000
--------------------------------------------------------------------------------
Top result: D03_Water_Quality_Monitoring.txt | Score: 0.482761
================================================================================
```

```
================================================================================
                TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: supply chain inventory
Processed query: supply chain inventory
--------------------------------------------------------------------------------
RANK  DOCUMENT                                        COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D04_Supply_Chain_Optimization.txt                       0.299383
2     D08_Disaster_Response_Routing.txt                       0.062010
3     D02_Renewable_Wind_Energy.txt                           0.050349
4     D01_Cybersecurity_Threat_Detection.txt                  0.000000
5     D03_Water_Quality_Monitoring.txt                        0.000000
6     D05_Speech_Emotion_Recognition.txt                      0.000000
7     D06_Plant_Disease_Classification.txt                    0.000000
8     D07_Electric_Vehicle_Battery_Health.txt                 0.000000
9     D09_Digital_Banking_Authentication.txt                  0.000000
10    D10_Air_Pollution_Forecasting.txt                       0.000000
--------------------------------------------------------------------------------
Top result: D04_Supply_Chain_Optimization.txt | Score: 0.299383
================================================================================
```

```
================================================================================
                TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: speech emotion
Processed query: speech emotion
--------------------------------------------------------------------------------
RANK  DOCUMENT                                    COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D05_Speech_Emotion_Recognition.txt                    0.241380
2     D01_Cybersecurity_Threat_Detection.txt                0.000000
3     D02_Renewable_Wind_Energy.txt                         0.000000
4     D03_Water_Quality_Monitoring.txt                      0.000000
5     D04_Supply_Chain_Optimization.txt                     0.000000
6     D06_Plant_Disease_Classification.txt                  0.000000
7     D07_Electric_Vehicle_Battery_Health.txt               0.000000
8     D08_Disaster_Response_Routing.txt                     0.000000
9     D09_Digital_Banking_Authentication.txt                0.000000
10    D10_Air_Pollution_Forecasting.txt                     0.000000
--------------------------------------------------------------------------------
Top result: D05_Speech_Emotion_Recognition.txt | Score: 0.241380
================================================================================
```

```
================================================================================
               TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: plant disease
Processed query: plant disease
--------------------------------------------------------------------------------
RANK  DOCUMENT                                    COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D06_Plant_Disease_Classification.txt                0.237189
2     D01_Cybersecurity_Threat_Detection.txt              0.000000
3     D02_Renewable_Wind_Energy.txt                       0.000000
4     D03_Water_Quality_Monitoring.txt                    0.000000
5     D04_Supply_Chain_Optimization.txt                   0.000000
6     D05_Speech_Emotion_Recognition.txt                  0.000000
7     D07_Electric_Vehicle_Battery_Health.txt             0.000000
8     D08_Disaster_Response_Routing.txt                   0.000000
9     D09_Digital_Banking_Authentication.txt              0.000000
10    D10_Air_Pollution_Forecasting.txt                   0.000000
--------------------------------------------------------------------------------
Top result: D06_Plant_Disease_Classification.txt | Score: 0.237189
================================================================================
```

```
================================================================================
                TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: electric vehicle battery
Processed query: electric vehicle battery
--------------------------------------------------------------------------------
RANK  DOCUMENT                                    COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D07_Electric_Vehicle_Battery_Health.txt                0.526287
2     D01_Cybersecurity_Threat_Detection.txt                 0.000000
3     D02_Renewable_Wind_Energy.txt                          0.000000
4     D03_Water_Quality_Monitoring.txt                       0.000000
5     D04_Supply_Chain_Optimization.txt                      0.000000
6     D05_Speech_Emotion_Recognition.txt                     0.000000
7     D06_Plant_Disease_Classification.txt                   0.000000
8     D08_Disaster_Response_Routing.txt                      0.000000
9     D09_Digital_Banking_Authentication.txt                 0.000000
10    D10_Air_Pollution_Forecasting.txt                      0.000000
--------------------------------------------------------------------------------
Top result: D07_Electric_Vehicle_Battery_Health.txt | Score: 0.526287
================================================================================
```

```
================================================================================
               TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: disaster response routing
Processed query: disaster response routing
--------------------------------------------------------------------------------
RANK  DOCUMENT                                    COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D08_Disaster_Response_Routing.txt                    0.295572
2     D01_Cybersecurity_Threat_Detection.txt               0.075115
3     D02_Renewable_Wind_Energy.txt                        0.000000
4     D03_Water_Quality_Monitoring.txt                     0.000000
5     D04_Supply_Chain_Optimization.txt                    0.000000
6     D05_Speech_Emotion_Recognition.txt                   0.000000
7     D06_Plant_Disease_Classification.txt                 0.000000
8     D07_Electric_Vehicle_Battery_Health.txt              0.000000
9     D09_Digital_Banking_Authentication.txt               0.000000
10    D10_Air_Pollution_Forecasting.txt                    0.000000
--------------------------------------------------------------------------------
Top result: D08_Disaster_Response_Routing.txt | Score: 0.295572
================================================================================
```

```
================================================================================
                TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: digital banking authentication
Processed query: digital banking authentication
--------------------------------------------------------------------------------
RANK  DOCUMENT                                    COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D09_Digital_Banking_Authentication.txt                0.283205
2     D01_Cybersecurity_Threat_Detection.txt                0.075115
3     D02_Renewable_Wind_Energy.txt                         0.000000
4     D03_Water_Quality_Monitoring.txt                      0.000000
5     D04_Supply_Chain_Optimization.txt                     0.000000
6     D05_Speech_Emotion_Recognition.txt                    0.000000
7     D06_Plant_Disease_Classification.txt                  0.000000
8     D07_Electric_Vehicle_Battery_Health.txt               0.000000
9     D08_Disaster_Response_Routing.txt                     0.000000
10    D10_Air_Pollution_Forecasting.txt                     0.000000
--------------------------------------------------------------------------------
Top result: D09_Digital_Banking_Authentication.txt | Score: 0.283205
================================================================================
```

```
================================================================================
                TF-IDF INFORMATION RETRIEVAL SYSTEM
================================================================================
Query: air pollution forecasting
Processed query: air pollution forecasting
--------------------------------------------------------------------------------
RANK  DOCUMENT                                    COSINE SIMILARITY
--------------------------------------------------------------------------------
1     D10_Air_Pollution_Forecasting.txt                    0.433795
2     D02_Renewable_Wind_Energy.txt                        0.131259
3     D03_Water_Quality_Monitoring.txt                     0.086559
4     D04_Supply_Chain_Optimization.txt                    0.073312
5     D01_Cybersecurity_Threat_Detection.txt               0.000000
6     D05_Speech_Emotion_Recognition.txt                   0.000000
7     D06_Plant_Disease_Classification.txt                 0.000000
8     D07_Electric_Vehicle_Battery_Health.txt              0.000000
9     D08_Disaster_Response_Routing.txt                    0.000000
10    D09_Digital_Banking_Authentication.txt               0.000000
--------------------------------------------------------------------------------
Top result: D10_Air_Pollution_Forecasting.txt | Score: 0.433795
================================================================================
```